

SIMATS ENGINEERING

Assessment Tool 2: Industry Problem Solving Task

Course Name: Cloud Computing and Big Data Analytics

Course Code: CSA1558

Course Outcome Covered

CO4: Analyze big data characteristics and challenges across domains including security and application aspects. (BL4)

Bloom's Taxonomy: Articulation (Level 4) | Question Count: 5 | Total Marks: 40 | Duration: 60 Minutes

Code of Conduct

I **DEENADAYALAN P (Reg. No. 192521135)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

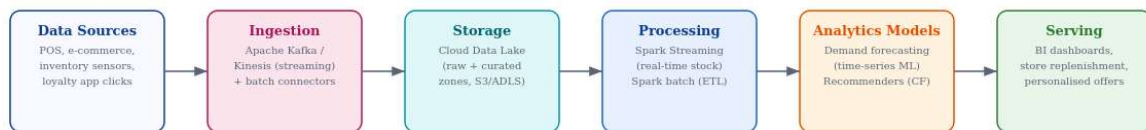
Signature of the student: P DEENADAYALAN

Q1. Retail Data Optimization Problem (8 Marks)

Understanding of Industry Context

The retailer runs both physical stores and an online platform, so the same product, customer, and order concepts appear in two very different data streams: structured POS transactions from stores and clickstream/order data from the website and app. The business goals stated are threefold — know stock levels as they change (real-time inventory), predict what will sell before it runs out (demand forecasting), and turn browsing/purchase history into personalised offers. This means the architecture has to support both fast, small updates (a single sale) and heavy analytical workloads (training a forecasting model on months of history) from the same underlying data.

Big Data Architecture



Store and online events stream through Kafka into a data lake; Spark handles both real-time inventory updates and batch model training, feeding demand-forecasting and recommendation models that drive dashboards and offers.

Fig 1: End-to-end big data pipeline for the retail chain

Tools, Technologies, and Analytics Models

Layer / Need	Tool or Technology	Why It Fits
Ingestion	Apache Kafka / AWS Kinesis	Captures POS sales and web clickstream events as they happen, decoupling fast-moving sources from downstream processing
Storage	Cloud data lake (S3/ADLS) with raw + curated zones	Cheaply retains raw history for retraining while curated tables serve fast queries
Real-time processing	Spark Structured Streaming	Updates live stock counts per store as soon as a sale event arrives
Batch processing	Spark batch / SQL warehouse (e.g. BigQuery, Redshift)	Aggregates historical sales for trend and seasonality analysis

Demand forecasting	Time-series models (ARIMA/Prophet) or gradient-boosted trees	Predicts short-term demand per SKU per store to guide replenishment
Personalisation	Collaborative filtering / recommendation engine	Matches customer purchase history and browsing behaviour to relevant offers

Security / Risk Awareness

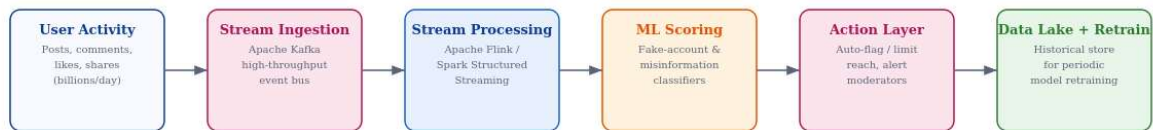
- Customer purchase and browsing data is personally identifiable — access to raw curated tables should be role-restricted and PII fields masked or tokenised before reaching the recommendation model
- Real-time inventory streams are business-critical: a Kafka outage or consumer lag could show incorrect stock, so the pipeline needs monitoring, replay capability, and graceful degradation to the last known good state
- Forecast and recommendation models should be periodically validated against actual sales to catch drift, so stale models do not silently drive poor stocking or irrelevant offers

Q2. Social Media Fraud Detection Problem (8 Marks)

Understanding of Industry Context

A platform processing billions of daily interactions cannot rely on periodic batch analysis to catch fake accounts or misinformation — by the time a nightly job finishes, harmful content may already have gone viral. The core challenge is therefore volume plus velocity: the system must inspect an enormous, continuous stream of posts, comments, likes, and shares and make a decision (flag, suppress, or allow) within seconds, not hours, while keeping false positives low enough that genuine users are not wrongly restricted.

Big Data Architecture



Every interaction flows through Kafka into Flink/Spark Streaming, which applies ML classifiers in near real time to score content and accounts; flagged items trigger automated action while all events are retained to continuously retrain the models.

Fig 2: Real-time fraud and misinformation detection pipeline

Role of Machine Learning and Stream Processing

Layer / Need	Tool or Technology	Why It Fits
Event bus	Apache Kafka	Absorbs bursts of billions of interactions per day without losing events, decoupling producers from consumers
Stream processing	Apache Flink / Spark Structured Streaming	Applies windowed aggregation (e.g., posting rate, account age vs. activity) in near real time
Fake-account detection	Graph-based ML (community/graph embeddings) + classification model	Fake accounts often form dense unusual clusters; graph features catch coordinated behaviour that content alone would miss

Misinformation detection	NLP classifiers (transformer-based text models)	Scores textual similarity to known false claims and flags rapidly-spreading suspicious content
Action layer	Rule engine + human moderation queue	Combines ML confidence scores with policy rules so borderline cases go to a human reviewer instead of being auto-removed

Security / Risk Awareness

- Automated flagging carries a free-speech and fairness risk — models must be monitored for bias across languages and demographics, with an appeal path for wrongly flagged users
- Adversaries adapt quickly, so the model pipeline needs continuous retraining on fresh labelled data rather than a static, one-time-trained classifier
- Interaction data at this scale is sensitive; access to raw user-level streams should be limited to the detection service, with aggregated/anonymised data used for any downstream analytics or reporting

Q3. Government Data Platform Problem (8 Marks)

Understanding of Industry Context

Unlike a single company's data platform, a national data platform must integrate datasets that were never designed to work together — census records, economic indicators, and geospatial layers — each owned and maintained by a different department with its own formats, update schedules, and legal constraints on disclosure. The central challenge is therefore less about processing speed and more about governance: making data interoperable and discoverable across agencies while strictly controlling who can see what, since some of this data (e.g., individual census responses) is highly sensitive.

Proposed Architecture

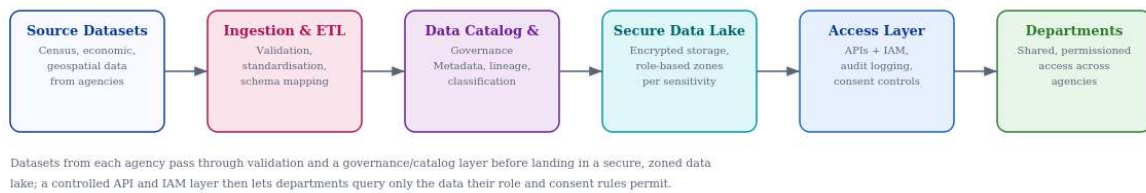


Fig 3: Governed, multi-agency data platform architecture

Governance and Data Management Strategy

Layer / Need	Tool or Technology	Why It Fits
Standardisation	Common data model + ETL validation layer	Ensures a "district" in census data matches the same boundary used in geospatial data before integration
Cataloguing	Data catalog with metadata & lineage tracking (e.g., Apache Atlas-style)	Lets departments discover what data exists and trace where a field originated, which is essential for trust in shared data
Storage	Secure, zoned data lake (public / restricted / classified)	Separates openly shareable statistics from sensitive record-level data by sensitivity tier
Access control	Federated IAM with role- and attribute-based access	Grants each department only the fields relevant to its mandate, honouring legal data-sharing agreements

Interoperability	Standard open APIs and data exchange formats	Allows agencies to consume shared data programmatically rather than through ad hoc file transfers
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Security / Risk Awareness

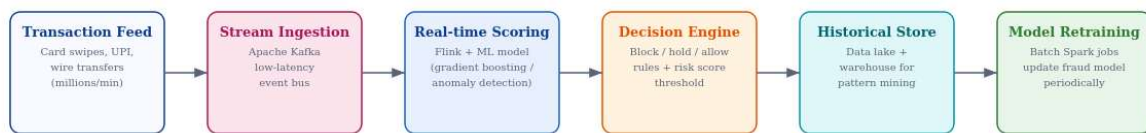
- Individual-level census and economic data must be de-identified or aggregated before cross-agency sharing to prevent re-identification when combined with geospatial data
- Every access to restricted data should be logged and auditable, since government platforms are high-value targets and are subject to public accountability
- A formal data governance board or policy (defining ownership, retention, and consent rules) is as important as the technical controls — access rules only work if departments agree on classification consistently

Q4. Banking Fraud Detection Problem (8 Marks)

Understanding of Industry Context

A bank processing millions of transactions per minute needs fraud detection that is both fast and accurate: too slow, and a fraudulent transaction completes before it can be stopped; too many false positives, and genuine customers get blocked, damaging trust. The system must also reconcile two timescales at once — scoring each transaction against a customer's long-term historical behaviour while reacting to it within milliseconds, which is a defining big data challenge of combining streaming and historical analysis.

Big Data Pipeline



Each transaction is scored within milliseconds by a streaming pipeline (Kafka + Flink) running a trained fraud model; high-risk transactions are held or blocked instantly, while all transactions feed a historical store used to retrain the model.

Fig 4: Real-time banking fraud detection pipeline

Technologies and Algorithms

Layer / Need	Tool or Technology	Why It Fits
Ingestion	Apache Kafka (low-latency partitioned topics)	Guarantees ordered, durable delivery of every transaction with sub-second latency
Real-time scoring	Apache Flink with an embedded/served ML model	Flink's low-latency stateful processing can maintain per-account behavioural features (e.g., rolling spend average) needed for scoring
Fraud model	Gradient-boosted trees (XGBoost) or autoencoder-based anomaly detection	Handles highly imbalanced fraud data well and can flag transactions that deviate from a customer's normal pattern
Decision engine	Rule engine combining model score + business rules	Lets the bank apply hard rules (e.g., blacklisted merchants) alongside the probabilistic model score
Historical analysis	Data lake + Spark batch jobs	Retrains the model regularly and supports investigator queries into past transaction patterns

Security / Risk Awareness

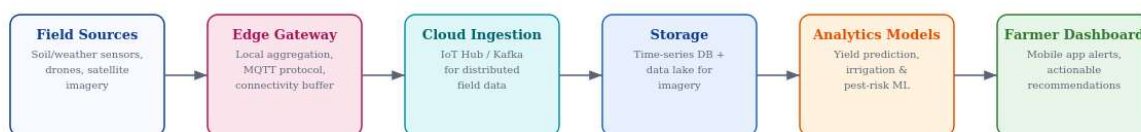
- The pipeline itself is a target: transaction data and model logic must be encrypted in transit and at rest, with strict network segmentation around the scoring service
- Regulatory compliance (e.g., PCI-DSS, RBI/central-bank guidelines) requires full auditability of every automated block/allow decision, so decisions and the features behind them must be logged, not just the final outcome
- Model drift is a real risk — fraud patterns evolve constantly, so accuracy must be monitored continuously and the model retrained on a defined cadence rather than left static

Q5. Agriculture Smart Farming Problem (8 Marks)

Understanding of Industry Context

Precision agriculture data originates from many small, distributed, often intermittently-connected sources — soil moisture sensors, weather stations, and drones flying over fields with patchy connectivity. Unlike a data centre workload, this system must tolerate unreliable networks at the edge and reconcile very different data types (numeric sensor readings versus drone imagery) into insights a farmer can act on quickly, since decisions like irrigation timing are time-sensitive.

Big Data Solution



Sensors and drones send field readings through an edge gateway to cloud ingestion; time-series and imagery data are stored and analysed by ML models that turn raw readings into yield, irrigation, and pest-risk recommendations for farmers.

Fig 5: Precision agriculture big data architecture

How Analytics Supports Decision-Making

Layer / Need	Tool or Technology	Why It Fits
Edge collection	Local gateway buffering + MQTT protocol	Tolerates intermittent field connectivity by buffering readings until a connection is available
Cloud ingestion	IoT Hub / Kafka	Aggregates readings from many distributed sensors and drones into a single ingestion point
Storage	Time-series database (sensor readings) + data lake (drone imagery)	Each data type is stored in the format best suited to how it will be queried later
Yield prediction	Regression / ensemble ML on soil, weather, and historical yield data	Gives farmers an early estimate of expected yield to plan logistics and sales
Irrigation & pest-risk models	Rule-based thresholds + ML classification on sensor/imagery patterns	Converts raw readings into a simple recommendation (e.g., irrigate now, inspect zone for pests)

Security / Risk Awareness

- Field devices are physically exposed and often low-power, so lightweight authentication (device certificates) is needed to stop spoofed sensor data from corrupting models
- Connectivity loss should degrade gracefully — cached local recommendations should still be available to the farmer even if the cloud link is temporarily down

- Farm-level data (yield, exact field locations) can be commercially sensitive; access should be scoped so aggregated regional insights can be shared without exposing an individual farmer's raw data

Rubric Self-Check

Each response above states the real-world context and constraints drawn from the scenario (Understanding of Industry Context), names specific big data tools mapped to a clear purpose (Application of Big Data Concepts), presents a scalable end-to-end architecture with a supporting diagram (Solution Design), lists concrete security and operational risks specific to that domain (Security / Risk Awareness), and follows a consistent structure of headings, tables, and diagrams throughout (Clarity and Professionalism).